

Outline

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- Lit Review
- Narrow down variables to more relevant ones
 - Remove duration, danceability
 - Add: Timbre, harmony, and register
- Using our own data
- Making conclusion about similarity between human music and between AI music
 - Control to Spotify
 - Control to AI
 - Spotify to AI

Features: BPM, timbre, harmony

Analysis

- Instrumental (Qualitative comparison)
 - Compare MFCCs
 - parametrize timbres
 - These also create a similarity
 - Timbre independent of pitch, need to use first 10 coefficients
 - Compare tempograms
 - parametrize beat and tempo info
 - Compare chromagrams
 - melodic features like pitch
 - Take short snapshots of the graphs and compare them for each song
- Cosine similarity → kNN
 - Comparing values in the MFCCs
 - More quantitative

Evaluation (need one of some kind)

- ROC

Deliverable

- Website
 - Interactive with song comparison
 - Subsection with our findings

Task Table

Week	Objective	Merina	Graham
4	Finish lit review Find new dataset	Extract these features from existing datasets	Lit review Become familiar with audio analysis techniques
5	Start analysis (split between Merina and Graham)		Exploratory data analysis Generate visualizations
6	Analysis Evaluation	Compare visualizations Finalize analysis methods Start poster	Evaluation metrics
7	Start deliverable	Continue analysis Evaluation metrics	Make website boilerplate (plan: use JS and Google Firebase and/or learn how to host a website via Github)
8	Deliverable	Update poster Write up literature review section of website	Finish website interactivity with JS Write up overview, short paper section of website
9	Paper?/other method of delivering analysis	Finish poster Finish summary	Finish website

Resources

Documentation

<https://librosa.org/doc/latest/index.html>

Papers

SVM for classifying music

<https://www.ee.columbia.edu/~sfchang/course/spr/papers/guo-li-svm-audio00.pdf>

Section 3.2 provides kind of a roadmap for analysis

<https://dl.acm.org/doi/10.1145/3769106>

Broader paper for our outlined process

<https://arxiv.org/html/2411.00195v1>

Computing similarity between songs, also talks about utilizing normal distributions and clustering

https://www.researchgate.net/profile/Francois-Pachet/publication/2909604_Finding_Songs_That_Sound_The_Same/links/0deec51eb9e4ca0c30000000/Finding-Songs-That-Sound-The-Same.pdf

Finding similarity between songs, but by using human subjects too

<https://archives.ismir.net/ismir2008/paper/000265.pdf>

Separating frequencies in audio, feature extraction

https://www.researchgate.net/publication/224150511_Music_mood_classification_by_rhythm_and_bass-line_unit_pattern_analysis?enrichId=rgreq-458165372a9237b62b42f9cf38144ff8-XXX&enrichSource=Y292ZXJQYWdlOzlyNDE1MDUxMTtBUzoxMDQ0NjYzMMA4MDEwMjVAMTQwMTkxODExNTA5OA%3D%3D&el=1_x_3

Sentiment analysis in music by tags or lyrics

<https://ismir2010.ismir.net/proceedings/ismir2010-45.pdf>

YouTube videos

Audio Processing with Python - Programming with Me, Helpful series

<https://youtu.be/PnGucYzeX7s?si=0ANuybDZPPTbytJg>

Textbook

If you want context, could be helpful for the research and findings section of the website or discussing limitations of our project

Ch 3-4 and Ch 6-7 are most relevant

<https://acrobat.adobe.com/id/urn:aaid:sc:US:99b1c89f-09fd-41c9-b3f6-e073bd562cfa>

Additional

General information and programming guidance on audio processing and analysis

<https://musicinformationretrieval.com/>

Explanation for MFCCs and chromagram relevance

<https://milvus.io/ai-quick-reference/how-is-similarity-measured-between-different-audio-clips>

Similar to how we're making our dataset with feature extraction

<https://medium.com/@alexandro.ramr777/audio-files-to-dataset-by-feature-extraction-with-librosa-d87adafe5b64>

Code notebook that describes the feature extraction

<https://www.kaggle.com/code/romeodavid/keras-librosa-mfcc-feature-extraction>

Further analysis ideas

<https://www.kaggle.com/code/adilshamim8/predicting-the-beats-per-minute-of-songs-101>

Reddit thread discussing techniques for music similarity

https://www.reddit.com/r/TheOverload/comments/1csqg0j/i_built_a_music_search_engine_to_help_you_find/

Interesting to think about

https://proceedings.neurips.cc/paper_files/paper/2023/file/7cbeec46f979618beafb4f46d8f39f36-Paper-Datasets_and_Benchmarks.pdf

Training existing model on our data

Using in conjunction with our visualizations

Their code is open source: <https://github.com/a43992899/MARBLE>

Glossary

Term	Definition
librosa	Python library for audio analysis
timbre	Tone quality
Cosine similarity	Used in ML approaches of analyzing song similarity, done between lyrics